Model for "MoS2"

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General Condition

- Basis type: `lg`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1']
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1', A_1'', A_2', A_2'', E', E'']
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1', A_1'', A_2', A_2'', E', E'']
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_1', A_1'', A_2', A_2'', E', E'']
- Max. neighbor: `10`
- Search cell range: `(-2, 3), (-2, 3), (-2, 3)`
- Toroidal priority: `false`

Group and Unit Cell

- Group: SG No. 187 D_{3h}^1 $P\bar{6}m2$ [hexagonal]
- Associated point group: PG No. 187 D_{3h} $\bar{6}m2$ (-6m2 setting) [hexagonal]
- Unit cell:
  $a =$ `3.16000`, $b =$ `3.16000`, $c =$ `20.00000`, $\alpha =$ `90.0`, $\beta =$ `90.0`, $\gamma =$ `120.0`
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ `[ 3.16000, 0.00000, 0.00000]`
  $\boldsymbol{a}_2 =$ `[-1.58000, 2.73664, 0.00000]`
  $\boldsymbol{a}_3 =$ `[ 0.00000, 0.00000,20.00000]`

Symmetry Operation

Table 1: Symmetry operation

	#	SO	#	SO	#	SO	#	SO	#	SO
	1	$\{1\|0\}$	2	$\{3^{+}_{001}\|0\}$	3	$\{3^{-}_{001}\|0\}$	4	$\{\mathrm{m}_{001}\|0\}$	5	$\{-6^{-}_{001}\|0\}$
	6	$\{-6^{+}_{001}\|0\}$	7	$\{\mathrm{m}_{110}\|0\}$	8	$\{\mathrm{m}_{100}\|0\}$	9	$\{\mathrm{m}_{010}\|0\}$	10	$\{2_{1-10}\|0\}$
	11	$\{2_{120}\|0\}$	12	$\{2_{210}\|0\}$						

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_1')$	A_1'	0	Q, T	-	-	1
2	$\mathbb{Q}_2(A_1')$	A_1'	2	Q, T	-	-	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
3	$\mathbb{Q}_3(A_1')$	A_1'	3	Q, T	-	-	$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$
4	$\mathbb{Q}_4(A_1')$	A_1'	4	Q, T	-	-	$\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$
5	$\mathbb{Q}_5(A_1')$	A_1'	5	Q, T	-	-	$-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$
6	$\mathbb{G}_2(A_1'')$	A_1''	2	G, M	-	-	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
7	$\mathbb{G}_3(A_1'')$	A_1''	3	G, M	-	-	$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$
8	$\mathbb{Q}_4(A_1'')$	A_1''	4	Q, T	-	-	$\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
9	$\mathbb{G}_1(A_2')$	A_2'	1	G, M	-	-	z
10	$\mathbb{G}_3(A_2')$	A_2'	3	G, M	-	-	$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$
11	$\mathbb{Q}_3(A_2')$	A_2'	3	Q, T	-	-	$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$
12	$\mathbb{Q}_1(A_2'')$	A_2''	1	Q, T	-	-	z
13	$\mathbb{G}_3(A_2'')$	A_2''	3	G, M	-	-	$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$
14	$\mathbb{Q}_3(A_2'')$	A_2''	3	Q, T	-	-	$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$
15	$\mathbb{Q}_4(A_2'')$	A_2''	4	Q, T	-	-	$\frac{3\sqrt{70}x^2yz}{4}-\frac{\sqrt{70}y^3z}{4}$
16	$\mathbb{Q}_{1,1}(E')$	E'	1	Q, T	-	1	x
17	$\mathbb{Q}_{1,2}(E')$					2	y
18	$\mathbb{G}_{2,1}(E')$	E'	2	G, M	-	1	$\sqrt{3}yz$
19	$\mathbb{G}_{2,2}(E')$					2	$-\sqrt{3}xz$
20	$\mathbb{Q}_{2,1}(E')$	E'	2	Q, T	-	1	$\sqrt{3}xy$
21	$\mathbb{Q}_{2,2}(E')$					2	$\frac{\sqrt{3}\left(x^2-y^2\right)}{2}$
22	$\mathbb{G}_{3,1}(E')$	E'	3	G, M	-	1	$\frac{\sqrt{15}z\left(-x^2+y^2\right)}{2}$
23	$\mathbb{G}_{3,2}(E')$					2	$\sqrt{15}xyz$
24	$\mathbb{Q}_{3,1}(E')$	E'	3	Q, T	-	1	$\frac{\sqrt{6}x\left(-x^2-y^2+4z^2\right)}{4}$
25	$\mathbb{Q}_{3,2}(E')$					2	$\frac{\sqrt{6}y\left(-x^2-y^2+4z^2\right)}{4}$
26	$\mathbb{Q}_{4,1}(E',1)$	E'	4	Q, T	1	1	$\frac{\sqrt{35}xy\left(-x^2+y^2\right)}{2}$
27	$\mathbb{Q}_{4,2}(E',1)$					2	$\frac{\sqrt{35}\left(x^4-6x^2y^2+y^4\right)}{8}$
28	$\mathbb{Q}_{4,1}(E',2)$	E'	4	Q, T	2	1	$\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$
29	$\mathbb{Q}_{4,2}(E',2)$					2	$\frac{\sqrt{5}\left(x^4-6x^2z^2-y^4+6y^2z^2\right)}{4}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
30	$\mathbb{G}_{1,1}(E'')$	E''	1	G, M	-	1	x
31	$\mathbb{G}_{1,2}(E'')$					2	y
32	$\mathbb{G}_{2,1}(E'')$	E''	2	G, M	-	1	$\sqrt{3}xy$
33	$\mathbb{G}_{2,2}(E'')$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
34	$\mathbb{Q}_{2,1}(E'')$	E''	2	Q, T	-	1	$\sqrt{3}yz$
35	$\mathbb{Q}_{2,2}(E'')$					2	$-\sqrt{3}xz$
36	$\mathbb{G}_{3,1}(E'')$	E''	3	G, M	-	1	$\frac{\sqrt{6}x(-x^2-y^2+4z^2)}{4}$
37	$\mathbb{G}_{3,2}(E'')$					2	$\frac{\sqrt{6}y(-x^2-y^2+4z^2)}{4}$
38	$\mathbb{Q}_{3,1}(E'')$	E''	3	Q, T	-	1	$\frac{\sqrt{15}z(-x^2+y^2)}{2}$
39	$\mathbb{Q}_{3,2}(E'')$					2	$\sqrt{15}xyz$
40	$\mathbb{Q}_{4,1}(E'')$	E''	4	Q, T	-	1	$\frac{\sqrt{10}yz(-3x^2-3y^2+4z^2)}{4}$
41	$\mathbb{Q}_{4,2}(E'')$					2	$\frac{\sqrt{10}xz(3x^2+3y^2-4z^2)}{4}$

Basis in full matrix

Table 3: dimension = 11

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$\|d_v\rangle$ @Mo(1)	1	$\|d_{xy}\rangle$ @Mo(1)	2	$\|d_{xz}\rangle$ @Mo(1)	3	$\|d_{yz}\rangle$ @Mo(1)	4	$\|d_u\rangle$ @Mo(1)
5	$\|p_x\rangle$ @S(1)	6	$\|p_y\rangle$ @S(1)	7	$\|p_z\rangle$ @S(1)	8	$\|p_x\rangle$ @S(2)	9	$\|p_y\rangle$ @S(2)

continued ...

Table 3

	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
	10	$\|p_z\rangle$ @S(2)								

Table 4: Atomic basis (orbital part only)

orbital	definition
$\|d_v\rangle$	$\frac{\sqrt{3}(x^2-y^2)}{2}$
$\|d_{xy}\rangle$	$\sqrt{3}xy$
$\|d_{xz}\rangle$	$\sqrt{3}xz$
$\|d_{yz}\rangle$	$\sqrt{3}yz$
$\|d_u\rangle$	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
$\|p_x\rangle$	x
$\|p_y\rangle$	y
$\|p_z\rangle$	z

SAMB: 30 (all 30)

- `Mo` : 'Mo' site-cluster
  * bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
  * ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$
  * wyckoff: `1a`

  z1 $\quad \mathbb{Q}_0^{(c)}(A_1') = \mathbb{Q}_0^{(a)}(A_1')\mathbb{Q}_0^{(s)}(A_1')$

z2 $\mathbb{Q}_{2}^{(c)}(A_1') = \mathbb{Q}_{2}^{(a)}(A_1')\mathbb{Q}_{0}^{(s)}(A_1')$

z3 $\mathbb{Q}_{4}^{(c)}(A_1') = \mathbb{Q}_{4}^{(a)}(A_1')\mathbb{Q}_{0}^{(s)}(A_1')$

- `S` : 'S' site-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `2h`

z4 $\mathbb{Q}_{0}^{(c)}(A_1') = \mathbb{Q}_{0}^{(a)}(A_1')\mathbb{Q}_{0}^{(s)}(A_1')$

z5 $\mathbb{Q}_{2}^{(c)}(A_1') = \mathbb{Q}_{2}^{(a)}(A_1')\mathbb{Q}_{0}^{(s)}(A_1')$

- `Mo;Mo_001_1` : 'Mo'-'Mo' bond-cluster
  * bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
  * ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$
  * wyckoff: `3b@3j`

z6 $\mathbb{Q}_{0}^{(c)}(A_1', a) = \mathbb{Q}_{0}^{(a)}(A_1')\mathbb{Q}_{0}^{(b)}(A_1')$

z7 $\mathbb{Q}_{0}^{(c)}(A_1', b) = \mathbb{M}_{1}^{(a)}(A_2')\mathbb{M}_{1}^{(b)}(A_2')$

z8 $\mathbb{Q}_{2}^{(c)}(A_1', a) = \mathbb{Q}_{2}^{(a)}(A_1')\mathbb{Q}_{0}^{(b)}(A_1')$

z9 $\mathbb{Q}_{2}^{(c)}(A_1', b) = \mathbb{M}_{3}^{(a)}(A_2')\mathbb{M}_{1}^{(b)}(A_2')$

z10 $$\mathbb{Q}_{3}^{(c)}(A_1', a) = \frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E')\mathbb{Q}_{1,1}^{(b)}(E') + \mathbb{Q}_{2,2}^{(a)}(E')\mathbb{Q}_{1,2}^{(b)}(E')\right)}{2}$$

z11 $$\mathbb{Q}_{3}^{(c)}(A_1', b) = \frac{\sqrt{58}\left(-2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E', 1)\mathbb{Q}_{1,1}^{(b)}(E') + \mathbb{Q}_{4,1}^{(a)}(E', 2)\mathbb{Q}_{1,1}^{(b)}(E') - 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E', 1)\mathbb{Q}_{1,2}^{(b)}(E') + \mathbb{Q}_{4,2}^{(a)}(E', 2)\mathbb{Q}_{1,2}^{(b)}(E')\right)}{58}$$

z12 $$\mathbb{Q}_{3}^{(c)}(A_1', c) = -\frac{\sqrt{2}\left(\mathbb{M}_{3,1}^{(a)}(E')\mathbb{T}_{1,1}^{(b)}(E', a) + \mathbb{M}_{3,2}^{(a)}(E')\mathbb{T}_{1,2}^{(b)}(E', a)\right)}{2}$$

z13 $\mathbb{Q}_{4}^{(c)}(A_1') = \mathbb{Q}_{4}^{(a)}(A_1')\mathbb{Q}_{0}^{(b)}(A_1')$

z14 $\mathbb{Q}_{5}^{(c)}(A_1') = -\frac{\sqrt{58}\left(\mathbb{Q}_{4,1}^{(a)}(E',1)\mathbb{Q}_{1,1}^{(b)}(E') + 2\sqrt{7}\mathbb{Q}_{4,1}^{(a)}(E',2)\mathbb{Q}_{1,1}^{(b)}(E') + \mathbb{Q}_{4,2}^{(a)}(E',1)\mathbb{Q}_{1,2}^{(b)}(E') + 2\sqrt{7}\mathbb{Q}_{4,2}^{(a)}(E',2)\mathbb{Q}_{1,2}^{(b)}(E')\right)}{58}$

- `S;Mo_001_1` : 'Mo'-'S' bond-cluster
  * bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `6a@6n`

z15 $\mathbb{Q}_{0}^{(c)}(A_1') = \frac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(a)}(E')\mathbb{Q}_{1,1}^{(b)}(E') + \mathbb{Q}_{1,2}^{(a)}(E')\mathbb{Q}_{1,2}^{(b)}(E') + \mathbb{Q}_{1}^{(a)}(A_2'')\mathbb{Q}_{1}^{(b)}(A_2'')\right)}{3}$

z16 $\mathbb{Q}_{2}^{(c)}(A_1', a) = \frac{\sqrt{7}\left(\sqrt{2}\mathbb{Q}_{3,1}^{(a)}(E')\mathbb{Q}_{1,1}^{(b)}(E') + \sqrt{2}\mathbb{Q}_{3,2}^{(a)}(E')\mathbb{Q}_{1,2}^{(b)}(E') + \sqrt{3}\mathbb{Q}_{3}^{(a)}(A_2'')\mathbb{Q}_{1}^{(b)}(A_2'')\right)}{7}$

z17 $\mathbb{Q}_{2}^{(c)}(A_1', b) = \frac{\sqrt{6}\left(-\mathbb{Q}_{1,1}^{(a)}(E')\mathbb{Q}_{1,1}^{(b)}(E') - \mathbb{Q}_{1,2}^{(a)}(E')\mathbb{Q}_{1,2}^{(b)}(E') + 2\mathbb{Q}_{1}^{(a)}(A_2'')\mathbb{Q}_{1}^{(b)}(A_2'')\right)}{6}$

z18 $\mathbb{Q}_{2}^{(c)}(A_1', c) = -\frac{\sqrt{2}\left(\mathbb{G}_{2,1}^{(a)}(E')\mathbb{Q}_{1,1}^{(b)}(E') + \mathbb{G}_{2,2}^{(a)}(E')\mathbb{Q}_{1,2}^{(b)}(E')\right)}{2}$

z19 $\mathbb{Q}_{3}^{(c)}(A_1', a) = \mathbb{Q}_{3}^{(a)}(A_1')\mathbb{Q}_{0}^{(b)}(A_1')$

z20 $\mathbb{Q}_{3}^{(c)}(A_1', b) = -\frac{\sqrt{2}\left(\mathbb{Q}_{3,1}^{(a)}(E'')\mathbb{Q}_{2,1}^{(b)}(E'') + \mathbb{Q}_{3,2}^{(a)}(E'')\mathbb{Q}_{2,2}^{(b)}(E'')\right)}{2}$

z21 $\mathbb{Q}_{3}^{(c)}(A_1', c) = \frac{\sqrt{2}\left(\mathbb{G}_{2,1}^{(a)}(E'')\mathbb{Q}_{2,1}^{(b)}(E'') + \mathbb{G}_{2,2}^{(a)}(E'')\mathbb{Q}_{2,2}^{(b)}(E'')\right)}{2}$

z22 $\mathbb{Q}_{4}^{(c)}(A_1') = \frac{\sqrt{7}\left(-\sqrt{6}\mathbb{Q}_{3,1}^{(a)}(E')\mathbb{Q}_{1,1}^{(b)}(E') - \sqrt{6}\mathbb{Q}_{3,2}^{(a)}(E')\mathbb{Q}_{1,2}^{(b)}(E') + 4\mathbb{Q}_{3}^{(a)}(A_2'')\mathbb{Q}_{1}^{(b)}(A_2'')\right)}{14}$

- `S;S_001_1` : 'S'-'S' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `1a@1c`

z23 $\mathbb{Q}_{0}^{(c)}(A_1') = \mathbb{Q}_{0}^{(a)}(A_1')\mathbb{Q}_{0}^{(b)}(A_1')$

z24 $\mathbb{Q}_2^{(c)}(A_1') = \mathbb{Q}_2^{(a)}(A_1')\mathbb{Q}_0^{(b)}(A_1')$

- `S;S_002_1` : 'S'-'S' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `6b@6n`

z25 $\mathbb{Q}_0^{(c)}(A_1', a) = \mathbb{Q}_0^{(a)}(A_1')\mathbb{Q}_0^{(b)}(A_1')$

z26 $$\mathbb{Q}_0^{(c)}(A_1', b) = \frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E'')\mathbb{Q}_{2,1}^{(b)}(E'') + \mathbb{Q}_{2,2}^{(a)}(E'')\mathbb{Q}_{2,2}^{(b)}(E'')\right)}{2}$$

z27 $$\mathbb{Q}_0^{(c)}(A_1', c) = \frac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(a)}(E'')\mathbb{M}_{1,1}^{(b)}(E'') + \mathbb{M}_{1,2}^{(a)}(E'')\mathbb{M}_{1,2}^{(b)}(E'') + \mathbb{M}_{1}^{(a)}(A_2')\mathbb{M}_{1}^{(b)}(A_2')\right)}{3}$$

z28 $\mathbb{Q}_2^{(c)}(A_1', a) = \mathbb{Q}_2^{(a)}(A_1')\mathbb{Q}_0^{(b)}(A_1')$

z29 $$\mathbb{Q}_2^{(c)}(A_1', b) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(a)}(E'')\mathbb{M}_{1,1}^{(b)}(E'') - \mathbb{M}_{1,2}^{(a)}(E'')\mathbb{M}_{1,2}^{(b)}(E'') + 2\mathbb{M}_{1}^{(a)}(A_2')\mathbb{M}_{1}^{(b)}(A_2')\right)}{6}$$

z30 $$\mathbb{Q}_3^{(c)}(A_1') = \frac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E')\mathbb{Q}_{1,1}^{(b)}(E') + \mathbb{Q}_{2,2}^{(a)}(E')\mathbb{Q}_{1,2}^{(b)}(E')\right)}{2}$$

Atomic SAMB

- bra: $\langle d_v|$, $\langle d_{xy}|$, $\langle d_{xz}|$, $\langle d_{yz}|$, $\langle d_u|$
- ket: $|d_v\rangle$, $|d_{xy}\rangle$, $|d_{xz}\rangle$, $|d_{yz}\rangle$, $|d_u\rangle$

x1 $$\mathbb{Q}_0^{(a)}(A_1') = \begin{bmatrix} \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \end{bmatrix}$$

x2 $\mathbb{Q}_2^{(a)}(A_1') = \begin{bmatrix} -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} \end{bmatrix}$

x3 $\mathbb{Q}_4^{(a)}(A_1') = \begin{bmatrix} \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{70}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{70}}{35} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{35} \end{bmatrix}$

x4 $\mathbb{Q}_4^{(a)}(A_1'') = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x5 $\mathbb{Q}_4^{(a)}(A_2'') = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x6 $\mathbb{Q}_{2,1}^{(a)}(E') = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{7} & 0 & 0 & 0 \end{bmatrix}$

x7 $\mathbb{Q}^{(a)}_{2,2}(E') = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \end{bmatrix}$

x8 $\mathbb{Q}^{(a)}_{4,1}(E', 1) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x9 $\mathbb{Q}^{(a)}_{4,2}(E', 1) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x10 $\mathbb{Q}^{(a)}_{4,1}(E', 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{7} & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 \end{bmatrix}$

x11 $\mathbb{Q}^{(a)}_{4,2}(E', 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{7} & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$

x12 $\mathbb{Q}^{(a)}_{2,1}(E'') = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \end{bmatrix}$

x13 $\mathbb{Q}^{(a)}_{2,2}(E'') = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\ 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \end{bmatrix}$

x14 $\mathbb{Q}^{(a)}_{4,1}(E'') = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{\sqrt{21}}{7} \\ 0 & 0 & 0 & \frac{\sqrt{21}}{7} & 0 \end{bmatrix}$

x15 $\mathbb{Q}^{(a)}_{4,2}(E'') = \begin{bmatrix} 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}}{7} \\ 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{7} & 0 & 0 \end{bmatrix}$

x16 $\mathbb{M}^{(a)}_{3}(A''_1) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x17 $\mathbb{M}_1^{(a)}(A_2') = \begin{bmatrix} 0 & -\frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x18 $\mathbb{M}_3^{(a)}(A_2') = \begin{bmatrix} 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{5} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x19 $\mathbb{M}_3^{(a)}(A_2'') = \begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & -\frac{i}{2} & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

x20 $\mathbb{M}_{3,1}^{(a)}(E') = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$

x21 $\mathbb{M}_{3,2}^{(a)}(E') = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

x22 $\mathbb{M}^{(a)}_{1,1}(E'') = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{10} & 0 \end{bmatrix}$

x23 $\mathbb{M}^{(a)}_{1,2}(E'') = \begin{bmatrix} 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & \frac{\sqrt{30}i}{10} \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{10} & 0 & 0 \end{bmatrix}$

x24 $\mathbb{M}^{(a)}_{3,1}(E'') = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 \end{bmatrix}$

x25 $\mathbb{M}^{(a)}_{3,2}(E'') = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 \\ \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 \end{bmatrix}$

- bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
- ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$

x26 $\mathbb{Q}^{(a)}_{0}(A'_1) = \begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

x27 $\mathbb{Q}_2^{(a)}(A_1') = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$

x28 $\mathbb{Q}_{2,1}^{(a)}(E') = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x29 $\mathbb{Q}_{2,2}^{(a)}(E') = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x30 $\mathbb{Q}_{2,1}^{(a)}(E'') = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

x31 $\mathbb{Q}_{2,2}^{(a)}(E'') = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$

x32 $\mathbb{M}_1^{(a)}(A_2') = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x33 $\mathbb{M}_{1,1}^{(a)}(E'') = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

x34 $\mathbb{M}_{1,2}^{(a)}(E'') = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

Cluster SAMB

- Site cluster

** Wyckoff: `1a`

y1 $\mathbb{Q}_{0}^{(s)}(A_1') = [1]$

** Wyckoff: `2h`

y2 $\mathbb{Q}_{0}^{(s)}(A_1') = \left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$

y3 $\mathbb{Q}_{1}^{(s)}(A_2'') = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$

- Bond cluster

** Wyckoff: `1a@1c`

y4 $\mathbb{Q}_{0}^{(b)}(A_1') = [1]$

y5 $\mathbb{T}_{1}^{(b)}(A_2'') = [i]$

** Wyckoff: `3b@3j`

y6 $\mathbb{Q}_{0}^{(b)}(A_1') = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

y7 $\mathbb{M}_{1}^{(b)}(A_2') = \left[-\frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{3}\right]$

y8 $\mathbb{Q}_{1,1}^{(b)}(E') = \left[\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}\right]$

y9 $\mathbb{Q}_{1,2}^{(b)}(E') = \left[-\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}\right]$

y10 $\mathbb{T}^{(b)}_{1,1}(E', a) = \left[-\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{3}, -\frac{\sqrt{6}i}{6}\right]$

y11 $\mathbb{T}^{(b)}_{1,2}(E', a) = \left[-\frac{\sqrt{2}i}{2}, 0, \frac{\sqrt{2}i}{2}\right]$

** Wyckoff: `6a@6n`

y12 $\mathbb{Q}^{(b)}_{0}(A'_1) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right]$

y13 $\mathbb{T}^{(b)}_{0}(A'_1) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}\right]$

y14 $\mathbb{Q}^{(b)}_{1}(A''_2) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

y15 $\mathbb{T}^{(b)}_{1}(A''_2) = \left[\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}\right]$

y16 $\mathbb{Q}^{(b)}_{1,1}(E') = \left[\frac{1}{2}, 0, -\frac{1}{2}, \frac{1}{2}, 0, -\frac{1}{2}\right]$

y17 $\mathbb{Q}^{(b)}_{1,2}(E') = \left[-\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}\right]$

y18 $\mathbb{T}^{(b)}_{1,1}(E', a) = \left[-\frac{i}{2}, 0, \frac{i}{2}, -\frac{i}{2}, 0, \frac{i}{2}\right]$

y19 $\mathbb{T}^{(b)}_{1,2}(E', a) = \left[\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}\right]$

y20 $\mathbb{M}^{(b)}_{1,1}(E'') = \left[\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}\right]$

y21 $\mathbb{M}^{(b)}_{1,2}(E'') = \left[\frac{i}{2}, 0, -\frac{i}{2}, -\frac{i}{2}, 0, \frac{i}{2}\right]$

y22 $\mathbb{Q}^{(b)}_{2,1}(E'') = \left[-\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{6}\right]$

y23 $\mathbb{Q}^{(b)}_{2,2}(E'') = \left[-\frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, 0, -\frac{1}{2}\right]$

** Wyckoff: `6b@6n`

y24 $\mathbb{Q}^{(b)}_{0}(A'_1) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}\right]$

y25 $\mathbb{M}^{(b)}_{2}(A''_1) = \left[-\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}\right]$

y26 $\mathbb{M}^{(b)}_{1}(A'_2) = \left[-\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}\right]$

y27 $\mathbb{Q}^{(b)}_{1}(A''_2) = \left[\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}\right]$

y28 $\mathbb{Q}^{(b)}_{1,1}(E') = \left[\frac{1}{2}, 0, -\frac{1}{2}, \frac{1}{2}, 0, -\frac{1}{2}\right]$

y29 $\mathbb{Q}^{(b)}_{1,2}(E') = \left[-\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}\right]$

y30 $\mathbb{T}^{(b)}_{1,1}(E', a) = \left[-\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}\right]$

y31 $\mathbb{T}^{(b)}_{1,2}(E', a) = \left[-\frac{i}{2}, 0, \frac{i}{2}, -\frac{i}{2}, 0, \frac{i}{2}\right]$

y32 $\mathbb{M}^{(b)}_{1,1}(E'') = \left[-\frac{i}{2}, 0, \frac{i}{2}, \frac{i}{2}, 0, -\frac{i}{2}\right]$

y33 $\mathbb{M}^{(b)}_{1,2}(E'') = \left[\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}\right]$

y34 $\mathbb{Q}^{(b)}_{2,1}(E'') = \left[-\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{6}\right]$

y35 $\mathbb{Q}^{(b)}_{2,2}(E'') = \left[-\frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, 0, -\frac{1}{2}\right]$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	Mo	$\|d_v\rangle$, $\|d_{xy}\rangle$, $\|d_{xz}\rangle$, $\|d_{yz}\rangle$, $\|d_u\rangle$
2	S	$\|p_x\rangle$, $\|p_y\rangle$, $\|p_z\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	Mo	Mo	[1]	[d]	[d]
2	Mo	S	[1]	[d]	[p]
3	S	S	[1,2]	[p]	[p]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'Mo' (#1) site cluster (1a), -6m2

SL	position ($\boldsymbol{s}$)	mapping
1	[0.00000, 0.00000, 0.00000]	[1,2,3,4,5,6,7,8,9,10,11,12]

Table 8: 'S' (#2) site cluster (2h), 3m.

SL	position ($\boldsymbol{s}$)	mapping
1	[0.33333, 0.66667, 0.07800]	[1,2,3,7,8,9]
2	[0.33333, 0.66667, 0.92200]	[4,5,6,10,11,12]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 9: 1-th 'Mo'-'Mo' [1] (#1) bond cluster (3b@3j), ND, $|\boldsymbol{v}|$= 3.16 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-1.00000,-1.00000, 0.00000]	[0.50000, 0.50000, 0.00000]	[1,4,-7,-10]	(1,1)	(1,1)	[1,1,0]
2	[1.00000, 0.00000, 0.00000]	[0.50000, 0.00000, 0.00000]	[2,5,-9,-12]	(1,1)	(1,1)	[-1,0,0]
3	[0.00000, 1.00000, 0.00000]	[0.00000, 0.50000, 0.00000]	[3,6,-8,-11]	(1,1)	(1,1)	[0,-1,0]

Table 10: 1-th 'Mo'-'S' [1] (#2) bond cluster (6a@6n), D, $|\boldsymbol{v}|$= 2.40044 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-0.33333, 0.33333,-0.07800]	[0.16667, 0.83333, 0.03900]	[1,7]	(1,1)	(1,1)	[0,-1,0]
2	[-0.33333,-0.66667,-0.07800]	[0.16667, 0.33333, 0.03900]	[2,9]	(1,1)	(1,1)	[0,0,0]
3	[0.66667, 0.33333,-0.07800]	[0.66667, 0.83333, 0.03900]	[3,8]	(1,1)	(1,1)	[-1,-1,0]
4	[-0.33333, 0.33333, 0.07800]	[0.16667, 0.83333, 0.96100]	[4,10]	(1,1)	(2,1)	[0,-1,-1]
5	[-0.33333,-0.66667, 0.07800]	[0.16667, 0.33333, 0.96100]	[5,12]	(1,1)	(2,1)	[0,0,-1]
6	[0.66667, 0.33333, 0.07800]	[0.66667, 0.83333, 0.96100]	[6,11]	(1,1)	(2,1)	[-1,-1,-1]

Table 11: 1-th 'S'-'S' [1] (#3) bond cluster (1a@1c), ND, $|\boldsymbol{v}|$= 3.12 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000, 0.00000,-0.15600]	[0.33333, 0.66667, 0.00000]	[1,2,3,-4,-5,-6,7,8,9,-10,-11,-12]	(2,1)	(1,1)	[0,0,1]

Table 12: 2-th 'S'-'S' [1] (#4) bond cluster (6b@6n), ND, $|\boldsymbol{v}|$= 3.16 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-1.00000,-1.00000, 0.00000]	[0.83333, 0.16667, 0.07800]	[1,-7]	(1,1)	(1,1)	[1,1,0]
2	[1.00000, 0.00000, 0.00000]	[0.83333, 0.66667, 0.07800]	[2,-9]	(1,1)	(1,1)	[-1,0,0]
3	[0.00000, 1.00000, 0.00000]	[0.33333, 0.16667, 0.07800]	[3,-8]	(1,1)	(1,1)	[0,-1,0]

continued ...

Table 12

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
4	[-1.00000,-1.00000, 0.00000]	[0.83333, 0.16667, 0.92200]	[4,-10]	(2,1)	(2,1)	[1,1,0]
5	[1.00000, 0.00000, 0.00000]	[0.83333, 0.66667, 0.92200]	[5,-12]	(2,1)	(2,1)	[-1,0,0]
6	[0.00000, 1.00000, 0.00000]	[0.33333, 0.16667, 0.92200]	[6,-11]	(2,1)	(2,1)	[0,-1,0]